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EDUCATION

Ph.D., Mechanical Engineering — Pohang University of Science and Technology (POSTECH) — Advisor: Prof. Junsuk Rho — Feb 2020 – Feb 2025

M.S., Mechanical Engineering — Pohang University of Science and Technology (POSTECH) — Advisor: Prof. Seongjin Park — Mar 2018 – Aug 2019

B.S., Mechanical Engineering — Pohang University of Science and Technology (POSTECH) — Mar 2013 – Feb 2018

EXPERIENCE

Assistant Professor — University of Science and Technology (UST), KIMS Campus, Advanced Materials Engineering — Sep 2026 – Present

Senior Researcher — Korea Institute of Materials Science (KIMS), Material Data & Analysis Research Division — Feb 2025 – Present

Technical Research Personnel (Alternative Military Service) — POSTECH — Feb 2022 – Feb 2025

Researcher — Korea Institute of Industrial Technology (KITECH), Intelligent Manufacturing (Root Technology) Research Institute — Neural network development for production and manufacturing — Aug 2019 – Dec 2019

Guest Researcher — Korea Institute of Industrial Technology (KITECH) — PI: Dr. Dongyoung Park — Aug 2018 – Dec 2018

FIRST-AUTHOR & CO-FIRST PUBLICATIONS

[1] K. Hur*, **C. Lee***, "Compressing the Validation Bottleneck: An Agentic Self-Driving Lab for Scientific Discovery", *arXiv preprint arXiv:2607.04508* (2026).

[2] **C. Lee**+, J. Rho*, "Benchmarking Optimization Methods Enabling Efficient Designs for Diverse Nanophotonic Applications", *Advanced Optical Materials*, Vol. 17, pp. 2500195 (2025).

[3] **C. Lee**+, D. Shin, Y. Kang, K. Choi, D. Park*, J. Rho*, "Real-Time Hot-Rolled Coil Placement Recommendation System with Data-Driven Model", *Advanced Intelligent Systems*, pp. 2400826 (2025).

[4] **C. Lee**+, S. Lee+, J. Seong, D. Park*, J. Rho*, "Inverse-designed metasurface for highly saturated transmissive colors", *JOSA B*, Vol. 41 (2024).

[5] J. Noh+, T. Badloe+, **C. Lee**+, J. Yun, S. So, J. Rho*, "Inverse design meets nanophotonics: From computational optimization to artificial neural network", *Intelligent Nanotechnology*, Vol. 3 (2023).

[6] **C. Lee**+, G. Chang+, J. Kim, G. Hyun, G. Bae, S. So, J. Yun, J. Seong, Y. Yang, D. Park, S. Jeon*, J. Rho*, "Concurrent Optimization of Diffraction Fields from Binary Phase Mask for Three-Dimensional Nanopatterning", *ACS Photonics*, Vol. 10, Issue 4 (2022).

[7] S. Moon+, **C. Lee**+, Y. Yang+, J. Kim, T. Badloe, C. Jung, C. Yoon, J. Rho*, "Tutorial on metalenses for advanced flat optics: design, fabrication, and critical considerations", *Journal of Applied Physics*, Vol. 131, Issue 9 (2022).

[8] D. Shin+, **C. Lee**+, U. Kühn, S. Lee, S. Park, H. Schwab, S. Scudino, K. Kosiba*, "Optimizing laser powder bed fusion of Ti-5Al-5V-5Mo-3Cr by artificial intelligence", *Journal of Alloys and Compounds*, Vol. 862, pp. 158018 (2021).

[9] **C. Lee**+, J. Na, K. Park, H. Yu, J. Kim, K. Choi, D. Park, S. Park, J. Rho*, S. Lee*, "Development of artificial neural network system to recommend process conditions of injection molding for various geometries", *Advanced Intelligent Systems*, Vol. 2, Issue 10 (2020).

[10] T. Lee+, **C. Lee**+, D. Oh, T. Badloe, J. Oh, J. Rho*, "Scalable and high-throughput top-down manufacturing of optical metasurfaces", *Sensors*, Vol. 20, Issue 15 (2020).

[11] D. Shin+, **C. Lee**+, S. Kim, D. Park, J. Oh, C. Gal, J. Koo, S. Park, S. Lee*, "Analysis of cold compaction for Fe-C, Fe-C-Cu powder design based on constitutive relation and artificial neural networks", *Powder Technology*, Vol. 353, Issue 15 (2019).

CO-AUTHOR PUBLICATIONS

[1] J. Seong, Y. Jeon, G. Han, S. Lee, Y. Yang, S. Kim, **C. Lee**, J. Kim, J. Kim, M. Jeong, J. Noh, J. An, J. Rho*, "Modal contrast engineering in ultraviolet and visible metalenses enabled by material-selective hybridization", *Nature Communications*, Vol. 17, pp. 7908 (2026).

[2] HC. Phan, MT. Tran, **C. Lee**, H. Kim, S. Oh, DK. Kim, HW. Lee*, "Parameter-aware high-fidelity microstructure generation using stable diffusion", *Advanced Engineering Informatics* (2026).

[3] J. Jung+, S. Oh, H. Kim, J. Na, SJ. Bae, **C. Lee**, SJ. Kim, HW. Lee*, "Multi-fidelity learning-based latent diffusion model for three-dimensional inverse microstructure design of dual phase steels", *Materials & Design*, Vol. 258, pp. 114623 (2025).

[4] J. Seong+, Y. Yang+, Y. Jeon, **C. Lee**, J. An, J. Rho*, "Structurally reordered crystalline atomic layer-dielectric hybrid metasurfaces for near-unity efficiency in the visible", *Materials Today*, Vol. 88, pp. 137 (2025).

[5] B. Ko+, J. Noh+, D. Chae, **C. Lee**, H. Lim, H. Lee, J. Rho*, "Neutral-Colored Transparent Radiative Cooler by Tailoring Solar Absorption with Punctured Bragg Reflectors", *Advanced Functional Materials*, pp. 2410613 (2024).

[6] H. Keawmuang+, T. Badloe+, **C. Lee**, J. Park, J. Rho*, "Inverse design of colored daytime radiative coolers using deep neural networks", *Solar Energy Materials and Solar Cells*, Vol. 271, pp. 112848 (2024).

[7] D. Park*, H. Lee, H. Song, KJ. Cha, SK. Hwang, **C. Lee**, "Artificial Neural Network-Based Process Recommender System for Additive Manufacturing", *TBD* (2024).

[8] HJ. Park, SM. Park, **C. Lee**, SK. Lee, J. Kim, J. Jang, CW. Park, D. Park*, "Quantitative Correlation between Brazed SUS304/MBF60 via Image Processing and Bonding Temperature", *TBD* (2024).

[9] C. Gal+, D. Shin, **C. Lee**, S. Park*, D. Park, "Rheological Behavior of Water-atomized 316L Stainless Steel Powder depending on Particle Size", *Metals and Materials International*, Vol. 29, pp. 3329-3339 (2023).

[10] S. So+, J. Kim+, T. Badloe, **C. Lee**, Y. Yang, H. Kang, J. Rho*, "Multicolor and 3D holography realized by inverse design of single-celled metasurfaces", *Advanced Materials*, pp. 2208520 (2023).

[11] J. Park, H. Keawmuang, T. Badloe, **C. Lee**, J. Rho*, "Utilizing Deep Neural Networks for the Inverse Design of Multilayered Daytime Radiative Coolers with Customizable Colors", *Materials Research Society (MRS) Proceedings* (2023).

[12] H. Cho+, J. Park, J. Kim, **C. Lee**, D. Park, J. Rho*, S. Park*, "Mass production of superhydrophilic micropatterned copper surfaces using powder injection molding process", *Powder Technology*, Vol. 411, pp. 117779 (2022).

[13] J. Noh+, Y. Nam+, S. So, **C. Lee**, S. Lee, Y. Kim, T. Kim, J. Lee, J. Rho*, "Design of a transmissive metasurface antenna using deep neural networks", *Optical Materials Express*, Vol. 11, Issue 7 (2021).

[14] D. Shin+, S. Kim, J. Oh, I. Jung, W. Yang, **C. Lee**, H. Kwon, J. Koo, S. Park*, "Correlation Study Between Material Parameters and Mechanical Properties of Iron–Carbon Compacts Using Sensitivity Analysis and Regression Model", *Metals and Materials International*, Vol. 25, pp. 1258 (2019).

*+ equal contribution · * corresponding author*

INDUSTRY & FUNDED PROJECTS

[1] **Development of a large-scale AI foundation model for metallic materials** — Korea Institute of Materials Science / Ministry of Science and ICT (Participating researcher — foundation model design and training). Jan 2026 – Dec 2030.

[2] **Construction of multimodal big data for metallic materials** — Korea Institute of Materials Science / Ministry of Science and ICT (Participating researcher — autonomous experimentation and multimodal data pipeline). Jan 2026 – Dec 2030.

[3] **Materials research innovation technology using multimodal artificial intelligence** — Korea Institute of Materials Science / Ministry of Science and ICT (Participating researcher — multimodal property prediction models). Jan 2025 – Dec 2027.

[4] **Planning of national mission-oriented convergence R&D programmes** — Korea Institute of Materials Science / Ministry of Science and ICT (Participating researcher — AI and materials programme planning). Jan 2025 – Dec 2027.

[5] **IRIS: construction of 3D printing materials data for aerospace applications (Phase 1)** — Korea Institute of Materials Science / Ministry of Trade, Industry and Energy (Participating researcher — generative microstructure models for PBF Ti-6Al-4V). Jan 2025 – Dec 2027.

[6] **K-PASS: digital innovation platform for metal materials manufacturing (Stage 2)** — Korea Institute of Materials Science / Ministry of Trade, Industry and Energy (Participating researcher — process AI and data platform). Jan 2025 – Dec 2026.

[7] **High-efficiency functional metalens design and printing technology** — Samsung Display (Achromatic large-area metalens design). Mar 2023 – Jan 2025.

[8] **Metaphotonics-based flat optics research institute (industry–academia convergence institute)** — POSCO (Large-area metalens design). Jul 2022 – Jan 2025.

[9] **Development of an optical platform for commercialising smart anti-counterfeiting technology** — National Research Foundation of Korea (Metalens design). Jun 2022 – Jan 2025.

[10] **Design and development of high-refractive polymer-based visible light absorber metalens with large area** — Samsung Display (Metalens design). Apr 2022 – Mar 2024.

[11] **Hologram Printing & Encoding of Super-Depth-3D Pattern (HOPE)** — National Research Foundation of Korea (with Korea University) (Inverse design of meta-masks for 3D nanopatterning). Apr 2022 – Jan 2025.

[12] **Development of ultra high-resolution metasurface color filter for display** — LG Display (Hybrid color-filter computation and inverse design). Mar 2021 – Feb 2022.

[13] **Development of an optimization model based on numerical analysis to minimize the temperature deviation of coil in the three-row curving yard** — POSCO (Cooling analysis, ANN-based cooling prediction, and coil placement optimisation system). Apr 2020 – Feb 2021.

[14] **Development of metasurface structures for collimating and steering the luminance distribution of LED / OLED light sources** — LG Display (LED optical modelling and metalens design). Mar 2020 – Feb 2021.

[15] **Development of manufacturing-optimal AI algorithms for root industries (intelligent root technology with add-on modules)** — Korea Institute of Industrial Technology (KITECH) (Neural network design, optimisation, and on-site deployment (press-machine defect pre-warning system)). Feb 2020 – Jan 2025.

[16] **A study on the propagation transformation using X-band metasurface structure** — Hanwha Defense (Metasurface design). Jan 2020 – Dec 2021.

[17] **Development of an AI injection molding system with a condition hit rate of 60% or higher** — National Research Foundation of Korea (Neural network development and deployment on the injection molding machine). Jan 2020 – Dec 2020.

[18] **Development of coil temperature prediction technology in the hot-rolling finishing yard to improve the cold-rolling yield of 780DP steel** — POSCO / RIST (FEM/FDM thermal model development and coil temperature prediction). Apr 2019 – Feb 2020.

[19] **Development of AI injection molding system having a shorter mold set-up time with minimum 60% probability than an expert** — LS Mtron (AI code development — ANN and simulation–experiment transfer learning). Jan 2018 – Dec 2019.

INTERNATIONAL CONFERENCE PRESENTATIONS

[1] K. Hur, **C. Lee**, "Compressing the Validation Bottleneck: An Agentic Self-Driving Lab for Scientific Discovery", *ICML 2026 AI4Science Workshop*, Seoul, Korea (Jul 2026) — poster.

[2] **C. Lee**, J. Rho, "Inverse Design of Diffraction Fields with Metamask for Three-dimensional Nanopatterning", *Metamaterials 2024*, Crete, Greece (Sep 2024) — oral.

[3] **C. Lee**, G. Chang, G. Bae, S. Jeon, J. Rho, "Inverse Design of Manufacturable Free-Form Metasurface for Three Dimensional Nanopatterning", *Nano Korea 2023*, Korea (Jul 2023) — oral.

[4] **C. Lee**, G. Chang, S. Jeon, J. Rho, "Diffraction Fields Optimization for Proximity-Field Nanopatterning: Toward High Contrast 3D", *2023 Materials Research Society Spring Meeting*, San Francisco, USA (Apr 2023) — poster.

[5] **C. Lee**, J. Rho, "Organic Light-emitting Diode Control (OLED) based on Metasurface", *Nano Convergence Conference 2021 — 19th International Nanotech Symposium & Exhibition*, Korea (Jul 2021) — virtual poster.

[6] **C. Lee**, S. Park, "Development of AI systems for smart injection molding based on digital twin", *International Injection Moulding Conference (IIMC 2019)*, Aachen, Germany (Apr 2019) — oral.

[7] **C. Lee**, S. Yoon, D. Kim, S. Park, "Deep Neural Network-based approach for warpage and weight prediction of plastic injection molding", *PM World Congress (PM 2018)*, Beijing, China (Sep 2018) — oral.

[8] J. Han, C. Gal, **C. Lee**, J. Cho, S. Park, "Development of bio-inspired piezoelectric hair cell sensor using powder injection molding", *Sixth Asia International Symposium on Mechatronics (AISM 2017)*, Pohang, Korea (Sep 2017) — oral.

DOMESTIC CONFERENCE PRESENTATIONS

[1] **C. Lee**, J. Rho, "3D Proximity-Field Nanopatterning Control based on Particle Swarm Optimization", *KSME Micro/Nano 2022*, Korea (May 2022) — oral.

[2] **C. Lee**, J. Rho, "Reversed design of metasurface antenna based on artificial neural networks", *KSME Micro/Nano 2020*, Korea (Jul 2021) — virtual poster.

[3] H. Kwon, **C. Lee**, S. Park, "Artificial neural network based processing parameter optimization for injection molding", *KSDME Winter Conference 2018*, Korea (Dec 2018) — oral.

[4] **C. Lee**, H. Kwon, D. Kim, S. Park, "Development a quality prediction model using deep neural network for injection molding", *Korean Powder Metallurgy Institute (KPMI)*, Yeosu, Korea (Mar 2018) — oral.

PATENTS

[1] **Electric-field-controlled 3D proximity-field patterning mask design apparatus, mask, manufacturing method, and nanopatterning apparatus** — 10-2023-0135822, Korea (2023).

[2] **Manufacturing method of meta-optical device, meta-optical device, and optical apparatus including the meta-optical device** — 10-2023-0085299, Korea, assignee: Samsung Display (2023).

[3] **AI-based placement system for minimizing hot-rolled coil temperature deviation** — 10-2023-0133116, Korea, assignee: POSCO (2023).

[4] **AI-based injection molding system and method for generating molding conditions** — WO 2021/049848 A1; US 2022/0388215 A1 (granted 2022-12-08); EP 4029672 A4 (granted 2023-10-18), PCT / United States / European Patent Office (2021).

[5] **Anti-lock braking system (ABS) for bicycles** — 10-2017-0027222, Korea (2017).

SPECIALIZATIONS

Forward Simulation. RCWA (in-house MATLAB), FDTD (Lumerical, MEEP), FEM (COMSOL), Bulk optics (VirtualLab Fusion), Scalar diffraction (in-house MATLAB)

Learning & Optimization. PyTorch, TensorFlow, Searching, PSO, Genetic, CMA-ES, Adjoint method, Automatic differentiation, Gradient, DDQN, Simulated Annealing, Gaussian Process (Bayesian Optimization)

Manufacturing Applications. Injection molding, Hot-rolled coil placement, Punch press, Drilling, DED metal 3D printing, Powder injection molding, Powder manufacturing

Nanophotonics Applications. LED simulation and collimation, 3D patterning mask design, Color filter, Nano antenna, Achromatic metalens, Meta-hologram, Beam forming

ACTIVE COLLABORATIONS

Nanomaterials for Electronics and Optics Lab, Korea University — PI: Prof. Seokwoo Jeon. 3D nanopatterning using metasurface.
Leibniz IFW Dresden, Leibniz IFW Dresden — PI: Prof. Konrad Kosiba. Optimization of laser powder bed fusion.
IKV-AACHEN, RWTH Aachen — PI: Dr. Pascal Bibow. Optimization of injection molding.
EM & Intelligent Design, Hanyang University — PI: Prof. Haejun Chung. Achromatic metalens inverse design using adjoint method.

RESEARCH INTERESTS

AI for materials, AI for manufacturing, Metamaterials and nanophotonic inverse design, Autonomous experimentation for metals.

GOOGLE SCHOLAR METRICS

Total citations: **805** · h-index: **12** · i10-index: **13** · as of 2026-08-20.